AGENTIC-OPD: RECOVERING FORGOTTEN AGENT CAPABILITIES VIA ON-POLICY DISTILLATION FOR AGENTIC TASKS

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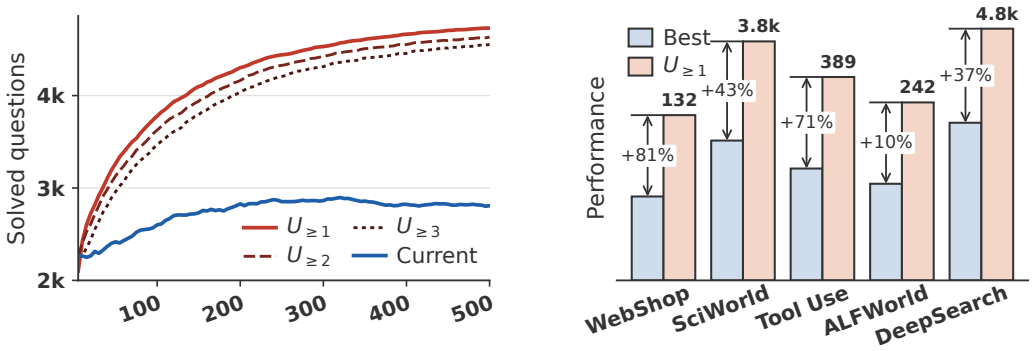
ABSTRACT

Reinforcement learning is now the mainstream paradigm for training LLMs on agentic tasks, yet standard practice keeps only the best-validation checkpoint and discards the rest. We show that this is lossy: in GRPO training of LLM agents, the best-validation checkpoint fails on problems that earlier checkpoints from the same run had already solved, and the union of problems solved across checkpoints far exceeds any single one. The discarded checkpoints thus form a teacher pool obtained at no additional training cost, which on-policy distillation (OPD) can mine to recover the lost coverage. Mining this pool raises two challenges: i) which checkpoint to distill from, as no checkpoint dominates the others and complementarity to the student is problem-specific; ii) where to trust it, as even a well-chosen teacher is only locally superior, so indiscriminate distillation overwrites behavior the student has already mastered. We propose Agentic-OPD, which treats the best-validation checkpoint as the student and all saved checkpoints as frozen teacher candidates, using the student's evolving capability gaps to guide both teacher selection and supervision filtering: the HCN policy (Highest Coverage Next) dynamically selects teachers by weighted coverage of the student's unsolved problems, while the LAD module (Locality-Aware Distillation) confines transfer to failed student rollouts on problems where the teacher performs better. Across five agentic task domains and 24 benchmarks and task subsets, Agentic-OPD improves over its initialization checkpoint by up to 7.4 points, with a 4B student surpassing the latest frontier models GPT-5.6 and DeepSeek-V4.1 on deep search. Agentic-OPD recasts a finished RL run as a reusable pool of policies and establishes post-RL capability recovery as a stage orthogonal to improvements in the RL algorithm itself.

1 INTRODUCTION

Reinforcement learning with verifiable outcome rewards has emerged as a prominent approach for turning language models into agents that can reason, call tools, and act over multiple turns (Xu et al., 2025; Li, 2025; Shao et al., 2024). Such runs produce dozens of checkpoints. Reporting only the best validation checkpoint leaves the capabilities of other checkpoints unmeasured. In agentic RL, an update can introduce a new behavior while erasing an earlier one. Across five agentic domains, we find that checkpoints from a single run collectively solve far more problems than any one checkpoint, revealing complementary capabilities along the training trajectory (Section 2).

Parameter-space merging produces one policy but can erase checkpoint-specific strengths, while separate inference increases deployment cost (Li et al., 2026b). Multi-teacher on-policy distillation (OPD) offers an alternative by consolidating these complementary capabilities into a single student policy (Agarwal et al., 2024; Gu et al., 2024). Prior work studies OPD with multiple teachers (Ma et al., 2026; Wang et al., 2026; Li et al., 2026a). Here, the teachers are temporal checkpoints from a single run, each of which may outperform the student on selected problems despite lower overall performance. This raises two challenges: **teacher selection**, choosing checkpoints that complement



(a) DeepSearch (HotpotQA)

(b) Best checkpoint vs. $U_{\geq 1}$

Figure 1: The temporal capability gap. **(a)** DeepSearch on HotpotQA: the current checkpoint against $U_{\geq k}(i)$, the tasks solved by at least k of the checkpoints through index i, for $k = 1, 2, 3$. **(b)** Best single checkpoint against the historical union $U_{\geq 1}$ in five domains, with the relative gap annotated.

the student, and **weak-to-strong distillation**, transferring locally useful behaviors from globally weaker checkpoints to the student.

We propose Agentic-OPD, which organizes distillation around the student's evolving capability gaps: whom to learn from and where to accept supervision. With all teacher checkpoints frozen, **HCN** (Highest Coverage Next) dynamically activates the candidate covering most of the student's current failures, while **LAD** (Locality-Aware Distillation) confines supervision to student-generated tokens in failed rollouts on problems where that teacher performs better. Across five agentic domains and 24 benchmarks and task subsets, Agentic-OPD delivers consistent gains over the converged GRPO initialization, achieving up to 7.4 points on DeepSearch across three backbones; with a 4B backbone it surpasses GPT-5.6 and DeepSeek-V4.1 in our DeepSearch evaluation.

Our contributions are:

- We quantify capability forgetting across five agentic domains, showing that checkpoints from a single RL run collectively solve far more problems than its best checkpoint.
- We formulate temporal checkpoint consolidation as weak-to-strong OPD: HCN dynamically routes teachers according to the student's capability gaps, while LAD transfers supervision only where a globally weaker teacher is locally stronger.
- We show that Agentic-OPD further improves performance after GRPO training has plateaued. We validate these gains across five agentic domains and on three backbones for DeepSearch.

2 Preliminary Study

An agent is a policy π that interacts with an external environment over multiple turns, interleaving reasoning, tool calls, and answer generation. We score only the outcome: for a task x with reference answer $y^*(x)$, let $\hat{y}_\pi(x)$ be its final answer and

$$C_\pi(x) = \mathbf{1}[\hat{y}_\pi(x) = y^*(x)]\,, \tag{1}$$

substituting the environment's own success predicate where the outcome is not an answer string.

Li et al. (2026b) report that CoT-style mathematical reasoning forgets intermediate abilities under both SFT and RL, with Temporal Forgetting Scores of 6.4% to 56.1%. We ask whether similar forgetting occurs in agentic reasoning, which involves multi-turn interaction with external environments.

We measure forgetting on the five domains of Section 4.1, training each from Qwen2.5-3B-Instruct with GRPO under result-based rewards and retaining every checkpoint of that run. Within each domain, all checkpoints share the same base model, data mixture, and tool environment. DeepSearch

is the detailed case: we train on HotpotQA for 500 steps and save and evaluate a checkpoint every 5 steps, yielding $n = 100$ checkpoints. The held-out validation set $\mathcal{D}_{\text{val}}$ consists of a random 10% split of the training queries. We use greedy decoding for all checkpoint evaluations. The other four domains follow the protocols of Section 4.1. Let $\mathcal{D}_{\text{val}}^{(d)}$ denote the validation set for domain d. Writing T_i for the i-th saved checkpoint, the solved set, the tasks solved by at least k of the first i checkpoints, and the gap on domain d are

$$Q_i^{(d)} = \left\{ x \in \mathcal{D}_{\text{val}}^{(d)} : C_{T_i}(x) = 1 \right\}, \tag{2}$$

$$U_{\geq k}^{(d)}(i) = \left\{ x \in \mathcal{D}_{\text{val}}^{(d)} : \textstyle\sum_{j=1}^{i} \mathbf{1}\left[x \in Q_j^{(d)} \right] \geq k \right\}, \qquad G_i^{(d)} = \left| U_{\geq 1}^{(d)}(i) \setminus Q_i^{(d)} \right|, \tag{3}$$

and we write $U_{\geq k}^{(d)} = U_{\geq k}^{(d)}(n)$ for the whole run. So $U_{\geq 1}^{(d)}(i) = \bigcup_{j \leq i} Q_j^{(d)}$ is the historical union through checkpoint i, and $G_i^{(d)}$ counts the tasks some checkpoint up to that point solves and T_i does not. Raising k drops the tasks only a single checkpoint ever solved, so $U_{\geq 2}^{(d)}$ and $U_{\geq 3}^{(d)}$ separate repeatedly solvable behavior from isolated successes.

The temporal capability gap persists across the evaluated checkpoints (Figure 1). On DeepSearch it widens along training, and in every domain the best checkpoint trails $U_{\geq 1}^{(d)}$ by the margin annotated in Figure 1(b); per-domain curves are in Appendix A.2. At step 500, the final checkpoint solves $2{,}808$ tasks, while $4{,}553$ tasks are solved by at least three checkpoints, a difference of 62%.

Historical checkpoints solve complementary subsets of tasks. This motivates selecting teachers by their coverage of the student's current failures.

3 AGENTIC-OPD

The checkpoint-level complementarity observed in Section 2 motivates Agentic-OPD. The student is initialized from the best-validation RL checkpoint and trained with OPD. Figure 2 illustrates the full pipeline: How to Build the Teacher Pool (a, Section 3.1), teacher scheduling via HCN (b, Section 3.2), supervision filtering via LAD (c, Section 3.3), and the consolidation loop that assembles the three (Section 3.4).

3.1 BUILD TEACHER POOL

Let $\{T_i\}_{i=1}^{n}$ be the checkpoints of one RL run in training-step order, instantiated with GRPO, in which every rollout is scored by the verifiable outcome indicator of Eq. 1. We write $r(x, \hat{y}) \in \{0, 1\}$ for that indicator applied to an individual rollout answer $\hat{y}$ rather than to a policy's decoded answer, so that $C_\pi(x) = r(x, \hat{y}_\pi(x))$.

Unless otherwise specified, we save checkpoints every 10 RL steps and freeze all saved checkpoints from the complete run into the teacher pool. The student is initialized from the best-validation RL checkpoint. HCN uses $\mathcal{D}_{\text{val}}$ to select teachers according to the problems they solve that the current student misses (Eq. 9).

3.2 HCN: WHICH TEACHER TO DISTILL FROM

With multiple frozen candidates in the pool, the order in which they teach the student matters. We propose **Highest Coverage Next** (HCN), a scheduling policy for activating them.

Gain matrix. Evaluating a policy on $\mathcal{D}_{\text{val}}$ splits it into the problems that policy solves and those it does not; a problem that a candidate solves and the student does not is a **gain** of that candidate. Consolidation proceeds in scheduling steps t, one every K training steps. Let $\pi_{S,t}$ denote the student policy at scheduling stage t. We record each candidate's solved set once and refresh the student's at each of them,

$$Q_i = \{ x \in \mathcal{D}_{\text{val}} : C_{T_i}(x) = 1 \}, \qquad i \in [n], \tag{4}$$

$$P_t = \left\{ x \in \mathcal{D}_{\text{val}} : C_{\pi_{S,t}}(x) = 1 \right\}, \tag{5}$$

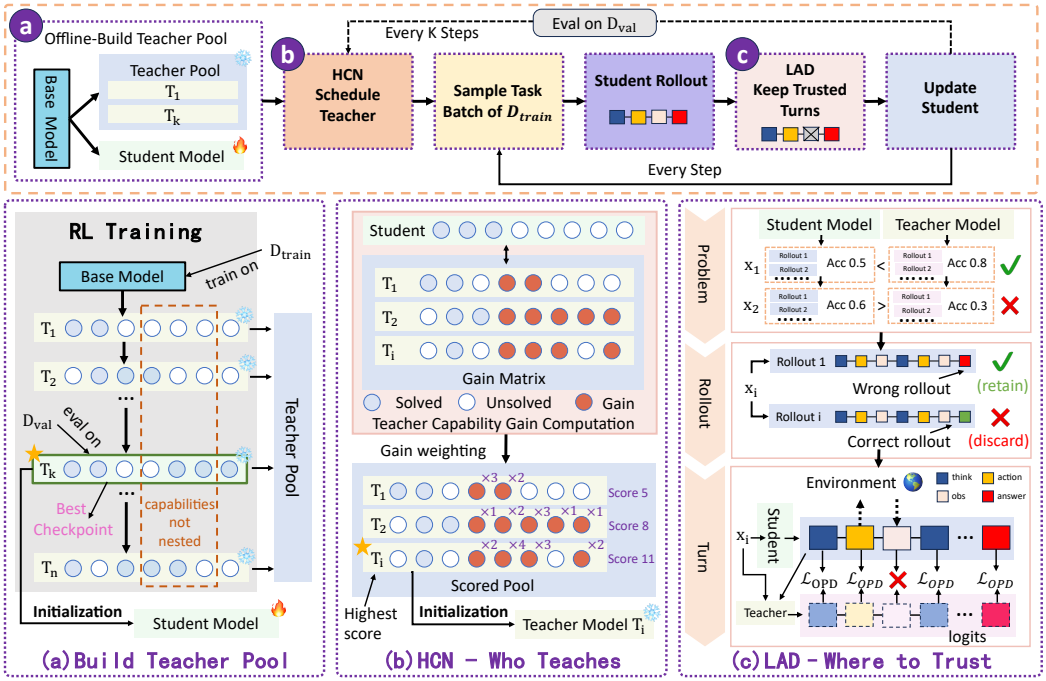


Figure 2: Overview of Agentic-OPD. **Top:** the consolidation loop. **(a)** Frozen checkpoints from one RL run form the teacher pool. **(b)** HCN selects teachers based on coverage of the student's unsolved tasks. **(c)** LAD filters supervision at the problem, rollout, and turn levels.

where HCN uses these sets to compute teacher scheduling scores. The gains of candidate i at scheduling step t are then

$$\Delta_{i,t} = Q_i \setminus P_t, \tag{6}$$

and stacking $\Delta_{1,t}, \ldots, \Delta_{n,t}$ as rows gives the gain matrix of Figure 2(b).

Scored pool. Each problem x carries a priority weight $w_t(x)$, initialized to $w_0(x) = 1$. The weighted coverage of candidate i is

$$g_{i,t} = \sum_{x \in \Delta_{i,t}} w_t(x). \tag{7}$$

After each scheduling step the weights are updated by

$$w_{t+1}(x) = \begin{cases} 0, & x \in P_{t+1}, \\ 1, & x \in Q_{i_t^*} \setminus P_{t+1}, \\ \min\{w_t(x) + \eta,\ w_{\max}\}, & \text{otherwise}, \end{cases} \tag{8}$$

with increment $\eta > 0$ and cap $w_{\max}$. Problems solved by the updated student receive zero weight. Unsolved problems covered by the current teacher return to base priority. The update increases the priority of problems that remain unsolved and are not covered by the current teacher, up to $w_{\max}$.

Scheduling teacher. HCN activates the candidate with the highest weighted coverage among those that still cover something the student misses,

$$T_t^* = T_{i_t^*}, \qquad i_t^* = \arg\max_{i \in \mathcal{C}_t} g_{i,t}, \qquad \mathcal{C}_t = \{i \in [n] : g_{i,t} > 0\}, \tag{9}$$

with ties broken toward the earlier checkpoint. The activated teacher serves as the supervisor for the next K training steps, after which the student is re-evaluated on $\mathcal{D}_{\text{val}}$, P_{t+1} is refreshed, and the process repeats. Consolidation stops when $\mathcal{C}_t = \emptyset$.

3.3 LAD: WHERE TO TRUST THE ACTIVATED TEACHER

A teacher selected by HCN covers problems the student currently misses, but its advantage may vary across training problems. We propose **Locality-Aware Distillation** (LAD), which selects supervision through filters at three levels shown top to bottom in Figure 2(c).

Problem level. For a problem $x \in \mathcal{D}_{\text{train}}$ the student and the activated teacher each draw G rollouts, both scored by the outcome reward r that the RL run already computes:

$$s(x) = \frac{1}{G}\sum_{j=1}^{G} r\Big(x, \hat{y}_S^{(j)}\Big), \qquad q_t(x) = \frac{1}{G}\sum_{j=1}^{G} r\Big(x, \hat{y}_{T_t^*}^{(j)}\Big). \tag{10}$$

We retain problems for which the teacher has a higher empirical success rate than the student,

$$\mathcal{X}_t = \{x : q_t(x) > s(x)\}, \tag{11}$$

which excludes problems where the teacher's empirical success rate does not exceed the student's, including those for which both rates are zero. In the top block of Figure 2(c), x_1 is admitted and x_2 is rejected because the student has a higher empirical success rate on it.

Rollout level. Inside an admitted problem only the rollouts the student failed carry a gap to close,

$$\mathcal{F}(x) = \Big\{j \in [G] : r\Big(x, \hat{y}_S^{(j)}\Big) = 0\Big\}, \qquad x \in \mathcal{X}_t. \tag{12}$$

The teacher's G rollouts estimate $q_t(x)$; distillation uses the student's failed trajectories. The middle block of Figure 2(c) retains the student's wrong rollout and discards the correct one.

Turn level. The frozen teacher runs a single forward pass over each trajectory τ retained by Eqs. 11 and 12. Because an agentic trajectory interleaves student turns with environment returns, the loss is computed over the student-generated positions $\mathcal{G}(\tau)$, comprising the think, action, and answer turns in the bottom block of Figure 2(c). Prompt tokens, observations, tool returns, and padding are excluded. At such a position ℓ we truncate to the student's top-k tokens $A_\ell^{(k)}$ and renormalize both policies on that support, $p_\pi^{(k)}(v \mid \tau_{<\ell}) = \pi(v \mid \tau_{<\ell}) / \sum_{u \in A_\ell^{(k)}} \pi(u \mid \tau_{<\ell})$, and minimize $D_{\text{KL}}(p_S^{(k)} \| p_T^{(k)})$

$$\mathcal{L}_{\text{OPD}} = \frac{1}{\sum_\tau |\mathcal{G}(\tau)|} \sum_\tau \sum_{\ell \in \mathcal{G}(\tau)} D_{\text{KL}}\Big(p_{S,t}^{(k)}(\cdot \mid \tau_{<\ell}) \;\Big\|\; p_{T_t^*}^{(k)}(\cdot \mid \tau_{<\ell})\Big). \tag{13}$$

3.4 THE CONSOLIDATION LOOP: HOW MUCH TO DRAW PER UPDATE

The three components compose into the loop at the top of Figure 2: HCN activates a teacher, the student rolls out problems drawn from $\mathcal{D}_{\text{train}}$, LAD selects the samples worth learning from, and the student is updated on them. Since LAD discards most of what is generated, we follow the oversampling of DAPO (Yu et al., 2026) and keep drawing batches of B problems until the sequences retained by Eqs. 11 and 12 fill one batch, discarding the surplus, with a draw capped at L batches. We use a fixed number of retained sequences per update and retain the optimizer hyperparameters of the unfiltered setting. Additional sampling and teacher evaluation incur inference cost (Appendix A.3).

4 EXPERIMENTS

To verify the stability and effectiveness of Agentic-OPD, we conduct extensive experiments across a wide range of scenarios and environments.

4.1 EXPERIMENTAL SETUP

Tasks and Metrics. We evaluate Agentic-OPD on DeepSearch, Tool-use, WebShop, SciWorld, and ALFWorld. DeepSearch uses a Wikipedia retriever and is scored by exact match on seven question

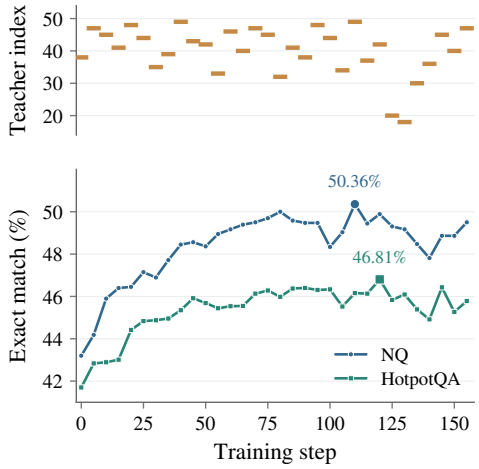
Table 1: DeepSearch exact match accuracy (%) on the test sets ($\mathcal{D}_{\text{test}}$) of seven benchmarks. **Bold** marks the best score within each backbone block.

	In-domain		Out-of-domain					
Model	NQ	HotpotQA	PopQA	TriviaQA	2Wiki	MuSiQue	Bamboogle	Avg.
Proprietary models								
GPT-5.6 (OpenAI, 2026)	7.7	22.2	32.2	56.5	17.4	11.2	30.4	25.4
Claude Sonnet 5 (Anthropic, 2026)	23.0	16.8	22.6	64.8	4.5	5.9	40.8	25.5
Grok-4.6 (xAI, 2026)	20.7	33.4	27.3	59.8	25.6	12.3	46.4	32.2
Open-source models, >100B parameters								
DeepSeek-V4.1 (Xu et al., 2026)	31.5	42.6	40.5	72.9	30.7	15.9	64.8	42.7
GLM-5.3-Flash (GLM-5 Team et al., 2026)	29.1	46.7	43.9	70.5	36.6	23.3	64.8	45.0
Qwen3.5-122B-A10B (Qwen Team, 2026a)	34.4	42.1	42.4	72.0	36.2	20.4	59.2	43.8
Open-source models, 20–100B parameters								
Llama-3.1-70B-Instruct (Grattafiori et al., 2024)	32.3	37.7	37.5	63.5	30.1	16.1	48.0	37.9
Qwen3.6-35B-A3B (Qwen Team, 2026b)	34.8	38.6	41.0	68.1	30.0	17.5	56.0	40.9
Qwen3.5-35B-A3B (Qwen Team, 2026a)	32.1	39.7	38.9	70.3	30.0	18.8	53.6	40.5
Qwen3.8-27B (Qwen Team, 2026c)	33.2	41.0	36.2	56.7	26.5	15.0	48.0	36.0
Qwen3.6-27B (Qwen Team, 2026b)	32.7	38.2	38.6	65.5	29.0	15.9	52.0	38.9
Qwen3.5-27B (Qwen Team, 2026a)	31.1	39.5	36.1	67.8	31.9	17.5	52.8	39.5
Open-source models, ≤ 20B parameters								
Qwen3.5-9B (Qwen Team, 2026a)	23.7	28.2	28.3	56.9	28.5	10.7	38.4	30.7
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	28.7	28.3	33.5	52.3	21.0	10.7	43.2	30.4
Backbone: Qwen2.5-3B-Instruct (Qwen Team, 2024)								
Base model	27.9	21.2	33.3	49.6	21.0	6.6	22.4	26.0
GRPO	42.1	34.8	39.2	55.5	**35.8**	13.9	28.0	35.6
OPSD	24.8	21.0	31.5	45.9	23.3	4.5	24.8	25.1
Agentic-OPD	**45.4**	**37.9**	**43.2**	**60.8**	34.3	**14.6**	**32.8**	**38.4**
Backbone: Qwen2.5-7B-Instruct (Qwen Team, 2024)								
Base model	26.0	28.3	26.8	53.0	25.9	12.4	35.2	29.6
GRPO	43.2	39.2	41.7	60.9	29.9	16.3	40.8	38.8
OPSD	27.4	29.4	27.3	52.8	26.3	12.9	32.8	29.9
Agentic-OPD	**50.4**	**46.5**	**47.8**	**66.5**	**41.1**	**22.5**	**48.8**	**46.2**
Backbone: Qwen3-4B-Instruct-2507 (Qwen Team, 2025)								
Base model	25.7	24.1	28.6	49.0	25.1	6.2	37.6	28.0
GRPO	50.4	49.2	48.6	66.4	45.3	22.6	52.0	47.8
OPSD	30.4	30.9	34.8	52.3	17.3	9.4	25.6	28.7
Agentic-OPD	**55.6**	**52.2**	**50.8**	**69.1**	**46.1**	**24.4**	**54.8**	**50.4**

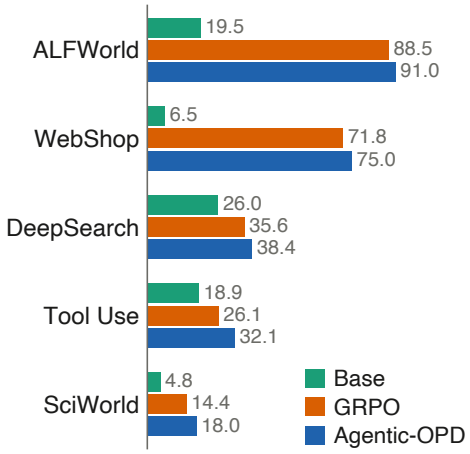
answering benchmarks. Tool-use provides a Python interpreter and is evaluated by pass@1 on AIME 2024, AIME 2025, MATH-500, AMC23, and OlympiadBench. We report task score and success rate for WebShop, Overall score for SciWorld, and success rate for ALFWorld. Appendix A.1 lists the benchmarks, interaction limits, and evaluation settings.

Models and Baselines. We use Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, and Qwen3-4B-Instruct for DeepSearch. The other four domains use Qwen2.5-3B-Instruct. All ablation and model-merging experiments use Qwen2.5-3B-Instruct. We compare with the instruction-tuned base model and GRPO. The DeepSearch comparison also includes OPSD (Zhao et al., 2026), which uses privileged information to construct teacher supervision. Table 1 additionally reports proprietary and open-source reference models evaluated without domain-specific training.

Training and Evaluation. RL and OPD use the same training data. The student is initialized from the best-validation checkpoint of each RL run. For DeepSearch, we use 90% of the NQ and HotpotQA training sets for training and reserve the remaining 10% as $\mathcal{D}_{\text{val}}$ for teacher selection and the preliminary capability analysis (Section 2). All results in Table 1 are measured on the corresponding benchmark test sets $\mathcal{D}_{\text{test}}$. RL training follows Search-R1 (Jin et al., 2025) for DeepSearch, ReTool (Feng et al., 2026) for Tool-use, and AgentGym-RL (Xi et al., 2025) for WebShop, SciWorld, and ALFWorld. Unless otherwise specified, we save a checkpoint every 10 RL steps and include all saved checkpoints from the full RL run in the teacher pool. OPD uses the same rollout sampling settings as RL and minimizes $D_{\text{KL}}(p_S^{(k)} \| p_T^{(k)})$ between the student and the selected teacher, as defined in Eq. 13. Evaluation follows the inference settings of the corresponding RL method. All training runs use one node with 8×A800 80GB GPUs.



(a) Teacher selection and QA performance. Base: Qwen2.5-7B-Instruct.



(b) Five-domain overview. Base: Qwen2.5-3B-Instruct.

Figure 3: **(a)** Teacher indices and NQ/HotpotQA exact match during consolidation; highlighted points mark peak accuracy. **(b)** Domain scores (%): mean exact match for DeepSearch, mean pass@1 for Tool-use, success rate for ALFWorld, mean of Score and Success for WebShop, and Overall score for SciWorld. Base denotes the instruction-tuned model.

4.2 Effectiveness and Generality Across Agentic Domains

Agentic-OPD improves mean DeepSearch accuracy over GRPO by 2.8, 7.4, and 2.6 percentage points on Qwen2.5-3B, Qwen2.5-7B, and Qwen3-4B, respectively (Table 1), with gains also extending to out-of-domain benchmarks. Historical checkpoints thus retain useful capabilities that the student can acquire through distillation after RL training.

With Qwen3-4B, Agentic-OPD reaches 50.4% mean accuracy, exceeding GPT-5.6 at 25.4% and DeepSeek-V4.1 at 42.7%. Under our DeepSearch evaluation settings, consolidating capabilities accumulated during task training allows a 4B model to outperform frontier models.

OPSD yields much smaller gains, suggesting that privileged information for a teacher using the same model as the student is insufficient for effective distillation in these agentic tasks. Agentic-OPD instead schedules historical teachers according to the student's current capability gaps and restricts distillation to supervision where the teacher has a local advantage. Figure 3(a) shows that teacher selection changes as the student develops and accuracy improves during consolidation, supporting selection based on complementary capabilities.

Across 24 benchmarks and task subsets spanning five agentic domains, Agentic-OPD improves aggregate performance over GRPO in every domain (Figure 3(b)). The gains extend beyond retrieval-based question answering to code execution and multi-step environment interaction, showing that historical checkpoint reuse can improve a single policy across different agentic tasks.

4.3 Capability Recovery and Teacher Complementarity

We examine the Qwen3-4B consolidation run on 2,000 routing-validation questions, with 1,000 each from NQ and HotpotQA. This analysis uses the 100 historical teachers saved every five RL steps through step 500 and the best observed OPD validation checkpoint, update 20. Since this checkpoint is selected on the same validation set, the analysis describes consolidation behavior rather than independent test-set performance.

Recovery and retention. Figure 4(a) separates gains from losses at the question level. The student recovers 60 questions from the historical union while losing 45 previously solved questions, for a net gain of 15 questions (+0.75 percentage points, from 55.25% to 56.0%). All newly solved

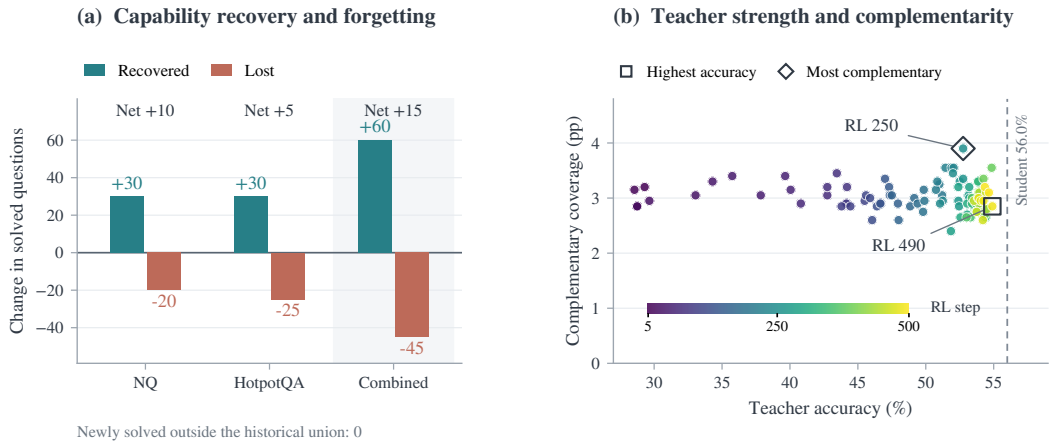


Figure 4: Capability recovery and teacher complementarity on the Qwen3-4B routing-validation set. **(a)** Question-level changes after OPD update 20. **(b)** Teacher accuracy versus unweighted coverage $100|Q_i \setminus P_{20}|/2{,}000$. Color indicates RL step; the dashed line marks student accuracy.

questions were solved by at least one historical teacher. Thus the observed improvement coincides with recovery of historical coverage, although substantial losses remain even at the best observed validation checkpoint.

Strength and complementarity. Figure 4(b) compares each teacher's overall accuracy with its coverage of the best student's remaining failures. The highest-accuracy teacher, RL step 490, scores 54.9% and covers 57 such questions (2.85 points). The most complementary teacher, step 250, has lower accuracy at 52.75% but covers 78 questions (3.9 points). Overall teacher accuracy therefore does not determine its usefulness for the student's remaining gaps, supporting the motivation for coverage-based scheduling.

4.4 Checkpoint Consolidation Analysis

Table 2 compares checkpoint reuse methods and ablations on NQ and HotpotQA with Qwen2.5-3B-Instruct.

Teacher scheduling and filtering. Replacing HCN with uniform teacher sampling at the same scheduling frequency reduces accuracy by 1.8 percentage points on both datasets. Removing LAD at the problem, rollout, and turn levels causes larger drops of 2.7 and 2.6 points on NQ and HotpotQA, respectively. Both variants still improve over GRPO, but lose much of the gain from consolidation. These results support selecting teachers for the student's remaining gaps and filtering their supervision according to local advantage.

Alternative reuse methods. We compare model soups (Wortsman et al., 2022), TIES and DARE-TIES (Yadav et al., 2023; Yu et al., 2024), checkpoint EMA (Sanyal et al., 2023), GTR-Turbo's alternating merging and RL updates (Wei et al., 2026), and Temporal Sampling (Li et al., 2026b). All methods use the same GRPO run; greedy soup has the same access to validation data as Agentic-OPD. Agentic-OPD achieves the highest accuracy on both datasets, improving over GRPO by 3.3 points on NQ and 3.1 points on

Table 2: Checkpoint reuse and ablations on DeepSearch with Qwen2.5-3B-Instruct: exact match (%).

Method	NQ	HotpotQA
GRPO (initialization)	42.1	34.8
Training-free merging		
Uniform soup	42.5	35.1
Greedy soup	43.4	35.9
TIES	43.0	35.6
DARE-TIES	43.7	36.2
EMA (last k)	42.8	35.5
Merging plus RL		
GTR-Turbo	45.0	37.4
Inference ensemble		
Temporal Sampling	44.6	37.1
Agentic-OPD	**45.4**	**37.9**
w/o HCN	43.6	36.1
w/o LAD	42.7	35.3

HotpotQA. DARE-TIES is the strongest merging method without further training, but trails Agentic-OPD by 1.7 points on each dataset. GTR-Turbo narrows the gap to 0.4 and 0.5 points with additional RL updates. Agentic-OPD also exceeds Temporal Sampling by 0.8 points on both datasets while using a single policy at inference. Thus, distilling selected historical teachers provides greater gains than the other reuse methods evaluated here.

5 Conclusion

The best checkpoint from an agentic RL run does not retain everything learned during training. Across five domains, historical checkpoints solve many tasks that the selected checkpoint misses. We use these checkpoints as frozen teachers to recover some of this lost coverage in a single student. HCN selects teachers according to the student's remaining failures, while LAD restricts distillation to problems where the teacher performs better and to failed student trajectories. The resulting policy improves aggregate performance over GRPO in all five domains, with gains across three backbones on DeepSearch. The ablations show why both decisions matter: removing either teacher scheduling or supervision filtering reduces the gains over the student's initialization on the evaluated NQ and HotpotQA benchmarks. These results support retaining historical checkpoints as a source of useful supervision after RL training. A checkpoint can still teach the student even when its overall accuracy is lower. Recovery remains incomplete, however, and some tasks show regressions. Consolidation also requires additional sampling and evaluation, although deployment uses only the student. Reducing this cost and measuring how much of the historical task coverage can be reliably recovered are useful next steps.

6 Related Work

6.1 Agentic RL and On-Policy Distillation

Verifiable-reward RL trains language models for multi-turn search, tool use, and other interactive tasks. Representative systems include Search-R1, s3, Agentic-R, ReTool, and AgentGym-RL (Jin et al., 2025; Jiang et al., 2025; Liu et al., 2026; Feng et al., 2026; Xi et al., 2025), with GRPO and its variants providing common optimization frameworks (Shao et al., 2024; Yu et al., 2026). These runs produce a sequence of checkpoints; Agentic-OPD treats that sequence as a source of complementary capabilities rather than using only one checkpoint.

Sequence-level distillation trains on teacher-generated trajectories, whereas on-policy distillation evaluates the student's own prefixes with teacher token distributions. GKD and MiniLLM established this paradigm, and recent work extends it to multi-turn agents through step-wise or turn-aware objectives (Agarwal et al., 2024; Gu et al., 2024; Zhong et al., 2026; Zhou et al., 2026). Multi-teacher methods study capability integration and conflicts among complementary teachers, including counteraction-aware and teacher-debate-based approaches (Ma et al., 2026; Chen et al., 2026; Wang et al., 2026; Men et al., 2026). Most use parallel or independently developed teachers. Agentic-OPD instead uses historical checkpoints as frozen temporal teachers: HCN selects them by the student's remaining capability gaps, and LAD restricts supervision to problems where the teacher has a local advantage and to failed student trajectories.

6.2 Historical Checkpoint Reuse and Model Merging

Model merging combines compatible models or checkpoints to aggregate capabilities. Model Soups averages weights from multiple fine-tuned models, while Task Arithmetic, Fisher Merging, TIES, and DARE improve parameter-space composition through task directions, parameter importance, interference resolution, and sparsification, respectively (Wortsman et al., 2022; Ilharco et al., 2022; Matena & Raffel, 2022; Yadav et al., 2023; Yu et al., 2024). More recent work extends merging to iterative model updating through activation-guided rotation-aware merging (Yao et al., 2026).

A related line of work reuses historical checkpoints from a single training trajectory. Checkpoint averaging can improve optimization stability and generalization (Sanyal et al., 2023; Li et al., 2026c), while Temporal Sampling recovers forgotten reasoning through outputs from multiple SFT or RL checkpoints (Li et al., 2026b). GTR-Turbo merges historical checkpoints into a unified teacher for

subsequent RL (Wei et al., 2026), and AgentPatch studies capability degradation after merging specialized agentic MLLMs and performs training-free capability repair (Shao et al., 2026). These approaches either reuse checkpoint outputs or collapse checkpoints into a single model. Agentic-OPD keeps them as separate frozen teachers and routes supervision according to the student's capability gaps and each teacher's local advantage, consolidating the recovered behavior into one deployable student.

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A APPENDIX

A.1 DETAILED ENVIRONMENT SETTINGS

RL training follows Search-R1 (Jin et al., 2025) for DeepSearch, ReTool (Feng et al., 2026) for Tool-use, and AgentGym-RL for WebShop, SciWorld, and ALFWorld. OPD uses the same training data and rollout sampling settings as RL. Unless otherwise specified, the teacher pool contains all checkpoints saved at intervals of 10 RL steps. OPD minimizes $D_{\mathrm{KL}}(p_S^{(k)} \| p_T^{(k)})$ in Eq. 13.

Table 3: Benchmarks, evaluation metrics, and maximum interaction turns per episode.

Domain	Benchmarks / tasks	Metric	Turns
DeepSearch	NQ (Kwiatkowski et al., 2019), HotpotQA (Yang et al., 2018) (in-domain); PopQA (Mallen et al., 2023), TriviaQA (Joshi et al., 2017), 2Wiki (Ho et al., 2020), MuSiQue (Trivedi et al., 2022), Bamboogle (Press et al., 2023) (out-of-domain)	Exact match	4
Tool-use	AIME 2024, AIME 2025, MATH-500 (Lightman et al., 2024), AMC23, OlympiadBench (He et al., 2024)	pass@1	16
WebShop (Yao et al., 2022)	Goal-directed shopping on simulated e-commerce pages	Score, success rate	20
SciWorld (Wang et al., 2022)	Measure, Test-Cond., Find, Chem-Mix, Lifespan	Overall score	15
ALFWorld (Shridhar et al., 2020)	Embodied household instruction following	Success rate	50

DeepSearch. The agent retrieves passages from a Wikipedia index through a search tool. We use 90% of the NQ and HotpotQA training sets for both RL and OPD and reserve the remaining 10% for teacher selection. Evaluation covers the seven benchmarks in Table 3 and follows the inference protocol of Search-R1 (Jin et al., 2025). Each episode allows up to 4 interactions.

Tool-use. The agent executes Python code in a sandboxed interpreter during reasoning. Training and evaluation follow ReTool (Feng et al., 2026), with at most 16 interaction turns per episode. For evaluation, we use temperature 1.0 and top-p 0.7, repeat evaluation 32 times, and report the average accuracy as an estimate of pass@1.

WebShop. The agent searches for products and navigates simulated shopping pages to complete a purchase instruction. Each episode allows up to 20 interaction turns. We report both task score and success rate.

SciWorld. The agent interacts with a text-based scientific environment. We report scores for Measure, Test-Cond., Find, Chem-Mix, and Lifespan, together with the Overall score. Each episode allows up to 15 interaction turns.

ALFWorld. The agent follows household instructions in a text-based environment. We evaluate Pick, Look, Clean, Heat, Cool, and Pick2, with at most 50 interaction turns per episode. We report success rates for each task type and for the full evaluation set.

For WebShop, SciWorld, and ALFWorld, RL training follows AgentGym-RL. Evaluation uses temperature 0.7, top-p 0.8, and top-k 20. Results are averaged over three random seeds.

A.2 TEMPORAL CAPABILITY GAP ACROSS DOMAINS

Figure 5 repeats the measurement of Section 2 on four further environments. For each domain we train with GRPO under result-based rewards, evaluate every saved checkpoint on the domain's

validation set, and plot the count solved by the current checkpoint, $|Q_i^{(d)}|$, against the historical union up to that step, $U_{\geq 1}^{(d)}(i)$; the shaded band is the gap $G_i^{(d)}$ of Eq. 3. Step 0 is the untrained checkpoint and is excluded, and the vertical axes are truncated below the lowest observed count.

The gap is present in every domain and does not close with training. It is widest on ReTool and WebShop, where the current checkpoint plateaus while $U_{\geq 1}^{(d)}(i)$ keeps growing, and narrowest on SciWorld, where both curves saturate near the size of the validation set and leave little room for the union to extend. This matches the DeepSearch observation: even within a single run, the latest checkpoint is not a superset of what earlier checkpoints could do, so the temporal pool retains complementary and recoverable behaviors.

Because SciWorld has a relatively small validation set, its point estimates have low statistical power and differences between checkpoints may fail to reach conventional significance thresholds even when their trajectories are visibly different. We therefore treat the SciWorld curves as descriptive evidence rather than as a standalone significance claim. For checkpoint comparisons, we use paired resampling over the same task instances (with a 95% bootstrap confidence interval) and aggregate estimates across the three random seeds; we do not interpret a non-significant SciWorld comparison as evidence of equivalence. The qualitative conclusion is retained only when the direction of the gap is consistent across seeds and is supported by the per-task trajectories.

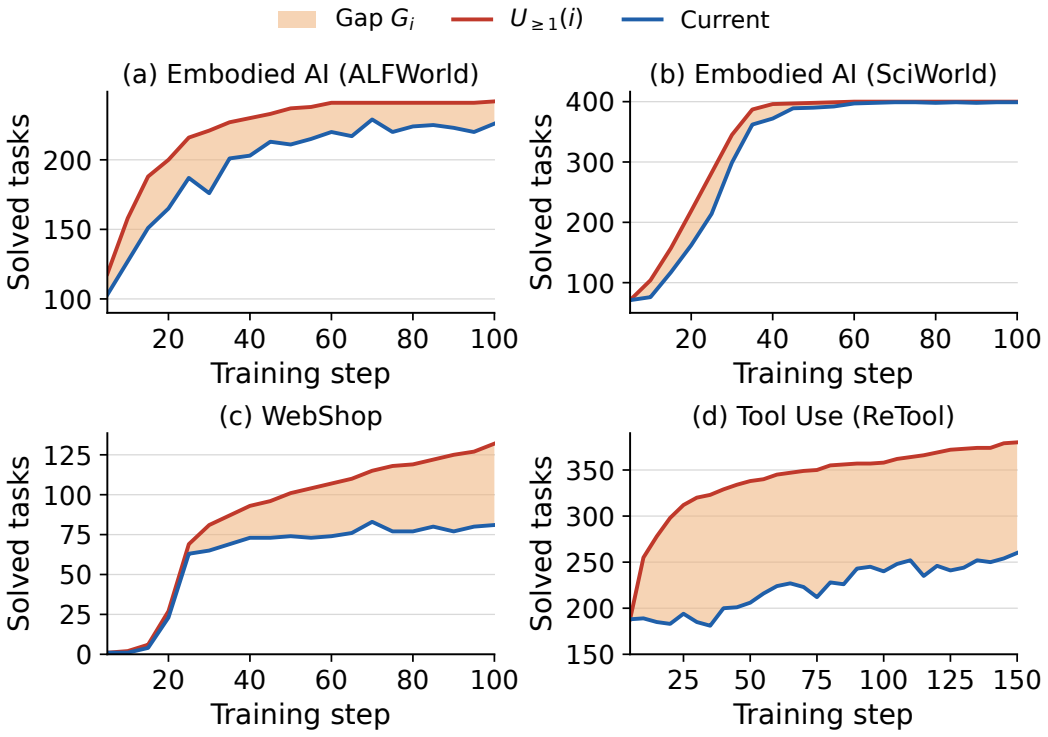


Figure 5: Validation tasks solved during RL in four domains: current checkpoint (blue) and historical union $U_{\geq 1}^{(d)}(i)$ (red). Shading marks the gap $G_i^{(d)}$.

Table 4: Tool-use pass@1 accuracy (%). Mean is the average across five benchmarks.

Method	AIME 2024	AIME 2025	MATH-500	AMC23	OlympiadBench	Mean
Base	3.0	3.1	43.0	27.6	17.9	18.9
GRPO	7.9	12.3	49.5	36.3	24.8	26.1
Agentic-OPD	**15.1**	**15.5**	**55.3**	**44.8**	**29.9**	**32.1**

Table 5: SciWorld scores by task subset and the aggregate Overall score.

Method	Measure	Test-Cond.	Find	Chem-Mix	Lifespan	Overall
Base	1.1	4.2	8.4	4.7	10.6	4.8
GRPO	7.6	9.3	22.9	15.4	41.1	14.4
Agentic-OPD	**14.8**	**10.0**	**25.0**	**26.6**	**50.0**	**18.0**

Table 6: WebShop task score and success rate (%). Mean averages the two metrics.

Method	Score	Success	Mean
Base	10.4	2.5	6.5
GRPO	80.6	63.0	71.8
Agentic-OPD	**84.0**	**66.0**	**75.0**

Table 7: ALFWorld success rate (%) by task type. All denotes the full evaluation set.

Method	Pick	Look	Clean	Heat	Cool	Pick2	All
Base	45.7	5.3	10.8	8.0	10.7	17.8	19.5
GRPO	97.8	73.7	**94.6**	**96.0**	82.1	80.0	88.5
Agentic-OPD	**100.0**	**78.9**	**94.6**	92.0	**85.7**	**86.7**	**91.0**

A.3 Cost of the Consolidation Loop

The consolidation loop of Section 3.4 targets a fixed number of retained sequences per update and retains the optimizer hyperparameters of the unfiltered OPD setting. For a target batch of $B_{\text{eff}} = BG$ sequences, if a fraction ρ of the student's rollouts survives the filters of Eq. 11–12, filling the batch requires approximately B/ρ sampled problems and $2BG/\rho$ rollouts, split evenly between the student and the activated teacher. The teacher also performs a forward pass over each retained student trajectory to provide token-level supervision. These estimates apply when the batch is filled before the cap of L sampling batches; updates that reach the cap may use fewer than B_{eff} retained sequences. Additional sampling, scoring, and teacher evaluation incur inference cost. HCN also requires evaluating the frozen teacher pool on the validation set and periodically re-evaluating the student for teacher scheduling.

A.4 Per-Benchmark Results

Figure 3(b) reports one aggregate score per domain: mean exact match across the seven DeepSearch benchmarks, mean pass@1 across the five Tool-use benchmarks, full-evaluation-set success rate (*All*) for ALFWorld, the mean of Score and Success for WebShop, and the environment's *Overall* score for SciWorld. DeepSearch per-benchmark results are reported in Table 1; the tables below give the per-benchmark and per-task results for the other four domains. All runs reported here start from Qwen2.5-3B-Instruct; "Base" denotes the instruction-tuned model before domain-specific training.

Agentic-OPD improves over GRPO on four of the six task types, ties on Clean, and trails on Heat. The aggregate *All* success rate increases from 88.5% to 91.0%, with the largest task-level gains on Look and Pick2.

Cross-benchmark analysis. On the tool-use benchmarks, the mean pass@1 increases from 26.1% with GRPO to 32.1% with Agentic-OPD. The method improves all five benchmarks. The largest changes are on AIME 2024 and AIME 2025 (from 3.0% to 15.1% and from 3.1% to 15.5%, respectively), whereas the gain on MATH-500 is smaller (49.5% to 55.3%).

The interactive tasks show a similar pattern. On SciWorld, the Overall score rises from 14.4 to 18.0, mainly due to gains on Chem-Mix and Lifespan; Test-Cond. changes only slightly. On WebShop, Score increases from 80.6% to 84.0% and Success from 63.0% to 66.0%, raising their mean from 71.8% to 75.0%.

ALFWorld also shows task-level variation: four task types improve, one is unchanged, and Heat drops from 96.0% to 92.0%. These results support an overall improvement, while also showing that the effect depends on the benchmark and task type.

A.5 FULL-PAGE TRAJECTORY CASES

Each case is rendered as a single self-contained image with a task panel and an Agentic-OPD trajectory panel. Reasoning is summarized for readability; retrieved passages are excerpted, with "[...]" marking omissions. Green highlights supporting evidence or successful actions, while red highlights errors. The two DeepSearch cases are validation rollouts collected at OPD update 20, rather than trajectories from an optimizer batch. They illustrate an entity-disambiguation failure and a successful two-hop search, respectively.

SciWorld

Task: Find the animal with the longest life span. The animals are in the 'outside' location. Focus on the animal with the longest life span.
Obs 0: This room is called the kitchen. In it, you see: the agent, a chair, a counter (bowl: apple, banana, orange, potato), a cupboard, a freezer, a fridge, a stopwatch (deactivated), a stove, a table... A door to the outside (that is closed).

Agentic-OPD

Thought 1: I need to find the door to the outside to reach the animals.
Act 1: go to door to the outside
Obs 1: The door is not open.

Thought 2: The door is closed, I need to open it to proceed.
Act 2: open door to the outside
Obs 2: The door is now open.

Thought 3: The door is open, I can go to the outside location.
Act 3: go to door to the outside
Obs 3: You move through the door to the outside.

Thought 4: Now I need to look for the animal with the longest life span.
Act 4: look around
Obs 4: This outside location is called the outside. Here you see: a baby beaver, a crocodile egg, a baby mouse, a fountain, an axe, a shovel...

Thought 5: I found the baby beaver and the crocodile egg. I need to examine them.
Act 5: pick up baby beaver / pick up crocodile egg
Obs 5: You move the crocodile to the inventory. You move the beaver to the inventory.

Thought 6: I examined both: the crocodile egg has a longer life span than the baby beaver, so the crocodile is the correct target.
Act 6: examine crocodile egg
Obs 6: a crocodile egg

Act 7-11: use stopwatch on crocodile egg -> "No known action matches that input"
Act 12-30: pick up orange -> "Unknown action" / "Ambiguous request" (repeats, never issues `focus on crocodile egg`)

Score: 50.0

Figure 6: **SciWorld case study.** The Agentic-OPD agent identifies the crocodile as the animal with the longest life span, but fails to issue the final focus action and receives partial reward.

WebShop

Instruction: Find me machine wash men's tuxedo shirts with polyester heathers, heathers cotton, cotton heather, needle sleeve, classic fit with color: heather blue, and fit type: men, and size: large, and price lower than 50.00 dollars.

Agentic-OPD

Thought 1: I should enter the search criteria to find the specific tuxedo shirts.
Act 1: search[machine wash men's tuxedo shirts ... color: heather blue, fit type: men, size: large, price lower than 50.00 dollars]
Obs 1: Page 1 (Total results: 50). [B07JVVDJ6L] Canadian Tuxedo T Shirt - Novelty Denim Tux T-Shirt - $19.99; [B07T3VMGDG] Tuxedo Cat 4th of July ... Raglan Baseball Tee - $23.99; ...

Thought 2: I should click the first product listed as it seems to match the criteria.
Act 2: click[B07JVVDJ6L]
Obs 2: fit type: men | women | youth ; color: royal blue | baby blue | heather grey | heather blue ; size: small | medium | large | x-large | xx-large | ... ; Canadian Tuxedo T Shirt - Novelty Denim Tux T-Shirt

Thought 3: I should select 'men' from the fit type to narrow down the results.
Act 3: click[men]
Obs 3: (product option page unchanged - fit type / color / size lists)

Thought 4: I should select 'heather blue' from the color options.
Act 4: click[heather blue]
Obs 4: (product option page unchanged - fit type / color / size lists)

Thought 5: I should select 'large' from the size options.
Act 5: click[large]
Obs 5: (product option page unchanged - fit type / color / size lists)

Thought 6: The product meets all the specified criteria; I should buy it now.
Act 6: click[Buy Now]

Score: 1.0

Figure 7: **WebShop case study.** The agent searches for the requested product, selects the required attributes, and completes the purchase.

ALFWorld

Task: put a clean tomato in sidetable.
Obs 0: You are in the middle of a room. Looking quickly around you, you see 4 cabinets, a coffeemachine, a countertop, diningtable 1-3, a drawer, a fridge, a garbagecan, a microwave, a sidetable, a sinkbasin, stoveburner 1-4, and a toaster.

Agentic-OPD

Thought 1: I need to find a tomato first. Let's start with the fridge.
Act 1: go to fridge 1
Obs 1: The fridge 1 is closed.

Act 2: open fridge 1
Obs 2: You open the fridge 1. In it, you see a apple 1, a tomato 4, a tomato 3, a tomato 2, and a tomato 1.

Act 3: take tomato 1 from fridge 1
Obs 3: You pick up the tomato 1 from the fridge 1.

Act 4: go to sinkbasin 1
Obs 4: On the sinkbasin 1, you see a butterknife 1, a lettuce 1, and a spatula 2.

Act 5: clean tomato 1 with sinkbasin 1
Obs 5: You clean the tomato 1 using the sinkbasin 1.

Act 6: go to sidetable 1
Obs 6: On the sidetable 1, you see a bowl 1, a peppershaker 2, a potato 1, and a spatula 1.

Act 7: put tomato 1 in/on sidetable 1

Score: 1.0

Figure 8: **ALFWorld case study.** The agent locates and cleans a tomato before placing it on the sidetable, completing the task.

MATH-500

Problem: Let n be a positive integer. What is the greatest possible value of gcd(n + 7, 2n + 1)? Give the final answer on its own line as \boxed{answer}.

Agentic-OPD

Reason: Recall GCD properties, then use the code interpreter to iterate over positive integers n and compute gcd(n + 7, 2n + 1), tracking the maximum.

Code (code_interpreter):

```
import math
max_gcd = 0
best_n = 0
for n in range(1, 10000): # Arbitrary upper limit, can be adjusted
    current_gcd = math.gcd(n + 7, 2 * n + 1)
    if current_gcd > max_gcd:
        max_gcd = current_gcd
        best_n = n
print(max_gcd)
```

Output: 13

Reason: The loop tracks the maximum gcd over n; the result 13 matches the manual calculation, so the greatest possible value is 13.

Answer: \boxed{13}

Score: 1.0

Figure 9: **MATH-500 case study.** The agent uses the code interpreter to verify the maximum GCD and returns $\boxed{13}$.

DeepSearch

Question: Who wrote and directed the film in which Adam Beach played Slipknot?
Ground truth: David Ayer
Setting: HotpotQA validation, OPD update 20; 2 searches.

Agentic-OPD

Thought: Identify the film in which Adam Beach played Slipknot, then find its writer and director.
Search 1: Adam Beach played Slipknot in which film
Obs 1: Doc 1, Slipknot (band): [...] In the meanwhile, Crahan made his directorial debut with the feature film ""Officer Downe"", based on a graphic novel of the same name and starring actor Kim Coates. [...] Doc 2, Adam Beach: [...] and played Slipknot in [passage ends]
Thought: The agent incorrectly links the incomplete Adam Beach passage to Officer Downe, mentioned in the band article.
Thought (raw): From the search result, it says: "Adam Beach played Slipknot in" — but the context is not fully clear. However, the only film mentioned where Adam Beach is associated with a character named "Slipknot" is in a film titled "Officer Downe", which is directed by Shawn "Clown" Crahan.
Search 2: Who directed the film Officer Downe starring Adam Beach
Obs 2: Doc 1, Officer Downe (film): Officer Downe is a 2016 film directed by Shawn Crahan and produced by Mark Neveldine. [...] Joe Casey, who created the comic series on which the film is based, wrote the film's screenplay.
Thought (raw): Since the film was directed by Shawn Crahan and written by Joe Casey, the answer must be the person who both wrote and directed it. However, the results show that only Crahan directed it, and Casey wrote it. Therefore, no single person both wrote and directed the film.
Thought: The agent notices that the director and writer differ, but retains the unsupported film identification and chooses its director.
Answer: Shawn Crahan

Score: 0.0

Figure 10: **DeepSearch case study (incorrect).** At OPD update 20, the agent confuses the character Slipknot with the band, follows an unsupported link to *Officer Downe*, and answers Shawn Crahan instead of David Ayer (score 0.0).

DeepSearch

Question: How many test centuries did the grandfather of Fabian Cowdrey make for England ?
Ground truth: 22
Setting: HotpotQA validation, OPD update 20; 2 searches.

Agentic-OPD

Thought: Identify Fabian Cowdrey's grandfather, then look up his number of Test centuries for England.
Search 1: who is the grandfather of Fabian Cowdrey
Obs 1: Doc 1, Fabian Cowdrey: [...] He made history by becoming the first third generation player to play for the county, following his father, Chris Cowdrey, and grandfather Colin Cowdrey. [...] Doc 2, Fabian Cowdrey: [...] His grandfather, Colin, Lord Cowdrey also captained both sides and played 114 Test matches for England, the first man to make 100 Test appearances. He was honoured for his services to cricket, the first English cricketer to be enobled in this way. [...]
Thought: The retrieved passage identifies Colin Cowdrey as the grandfather. Search for Colin's Test-century count.
Search 2: how many test centuries did Colin Cowdrey make for England
Obs 2: Doc 1, Colin Cowdrey: [...] Cowdrey made 22 Test Centuries to match Wally Hammond's England record [...] Of the 22 Tests in which Cowdrey made a hundred England won 10, drew 11 and lost 1, when he made 105 against South Africa at Trent Bridge in 1965. Cowdrey was the first man to score a Test century against six different nations; Australia, South Africa, the West Indies, New Zealand, India and Pakistan [...]
Thought: From the search results, Colin Cowdrey made 22 Test centuries for England.
Answer: 22

Score: 1.0

Figure 11: **DeepSearch case study (correct).** At OPD update 20, the agent identifies Colin Cowdrey as Fabian Cowdrey's grandfather, then retrieves his 22 Test centuries for England and returns the correct answer (score 1.0).